Microsoft Elevate

CAPSTONE PROJECT

STUDENT SOCIAL MEDIA ADDICTION ANALYSIS

PRESENTED BY

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PROBLEM STATEMENT:

Currently, social media platforms are widely used by students for communication, entertainment, and information sharing. While these platforms provide several benefits, excessive usage can affect students' academic performance, sleep patterns, mental health, and overall lifestyle. It is important to understand how students interact with social media and how usage patterns vary across different platforms and demographics. Eventually, gaining insights into students' social media habits becomes a key concern for educators and researchers. The crucial part is analyzing the relationship between social media usage, addiction levels, and academic performance to identify patterns and support better decision-making

PROPOSED SOLUTION:

To address the challenge of understanding students' social media usage and its impact on their academic performance, mental health, and lifestyle, a data-driven analytical approach is implemented. The solution involves collecting relevant data, processing it, and developing an interactive dashboard to visualize key insights.

Data Collection:

- A dataset containing information about students' demographics, social media usage patterns, addiction scores, sleep hours, mental health scores, relationship status, and academic performance impact is collected.
- This data provides the foundation for analyzing behavioral trends and identifying patterns in social media usage among students.

Data Preprocessing and Cleaning:

- The collected data is cleaned and prepared for analysis by handling missing values, verifying data consistency, and ensuring proper formatting of variables.

- This step ensures that the dataset is accurate and suitable for further analysis.

Data Analysis:

- The dataset is analyzed to identify relationships between key factors such as daily usage hours, addiction levels, mental health scores, and academic performance.
- Aggregations and calculations are performed to derive meaningful metrics and indicators.

Dashboard Development:

- An interactive dashboard is developed using Power BI to present the insights visually.
- The dashboard includes various visualizations such as bar charts, pie charts, maps, KPI indicators, and gauge charts to provide a comprehensive view of students' social media usage behavior.

Insight Generation:

- The dashboard enables users to explore patterns and trends through interactive filters and visualizations.
- This helps identify the most used platforms, average usage time, addiction levels, and the impact of social media on students' mental health and academic performance.

SYSTEM APPROACH:

The **System Approach** outlines the overall methodology used to analyze student social media usage and develop an interactive dashboard for identifying behavioral patterns and insights. The system focuses on collecting relevant data, processing it, and visualizing meaningful insights using analytical tools.

- **System Requirements**

To develop and analyze the student social media dataset, the following system requirements are needed:

Hardware Requirements

- Processor: Minimum Intel i3 or equivalent
- RAM: 4 GB or above
- Storage: Minimum 10 GB free space

Software Requirements

- Operating System: Windows 10 or above
- Data Visualization Tool: Power BI Desktop
- Data Source Format: Excel / CSV Dataset

- **Libraries / Tools Required**

The following tools and components are used to build the analytical dashboard:

- **Power BI Desktop** – Used for data visualization and dashboard development
- **Power Query** – Used for data transformation and cleaning
- **DAX (Data Analysis Expressions)** – Used for creating calculated measures and KPIs
- **Excel / CSV Dataset** – Used as the data source for analysis

These tools enable the system to process the dataset efficiently and generate interactive visualizations that help analyze social media usage patterns and their impact on students' academic performance and well-being.

REPORTS & ANALYSIS:

The **Reports and Analysis** section presents the key findings obtained from the student social media dataset using various visualizations and analytical metrics in the Power BI dashboard. The dashboard provides a comprehensive view of students' social media usage behavior and its relationship with mental health, academic performance, and lifestyle factors.

- **Social Media Platform Usage Analysis**

The analysis of social media platforms highlights the distribution of the most frequently used platforms among students. Platforms such as Instagram and TikTok show higher engagement compared to other platforms. This report helps in understanding which platforms are most dominant among students.

- **Daily Usage Analysis**

The report on average daily usage hours provides insights into the amount of time students spend on social media each day. Higher daily usage is often associated with increased addiction scores and reduced sleep hours, indicating a potential impact on students' daily routines.

- **Mental Health Analysis**

Mental health scores are analyzed across different social media platforms to understand how usage patterns may influence students' psychological well-being. The report helps identify whether higher usage levels correspond to lower mental health scores.

- **Addiction Level Analysis**

The addiction score analysis evaluates the level of social media dependency among students. By analyzing addiction scores across different academic levels and demographic groups, the dashboard highlights patterns of social media addiction.

- **Academic Performance Impact Analysis**

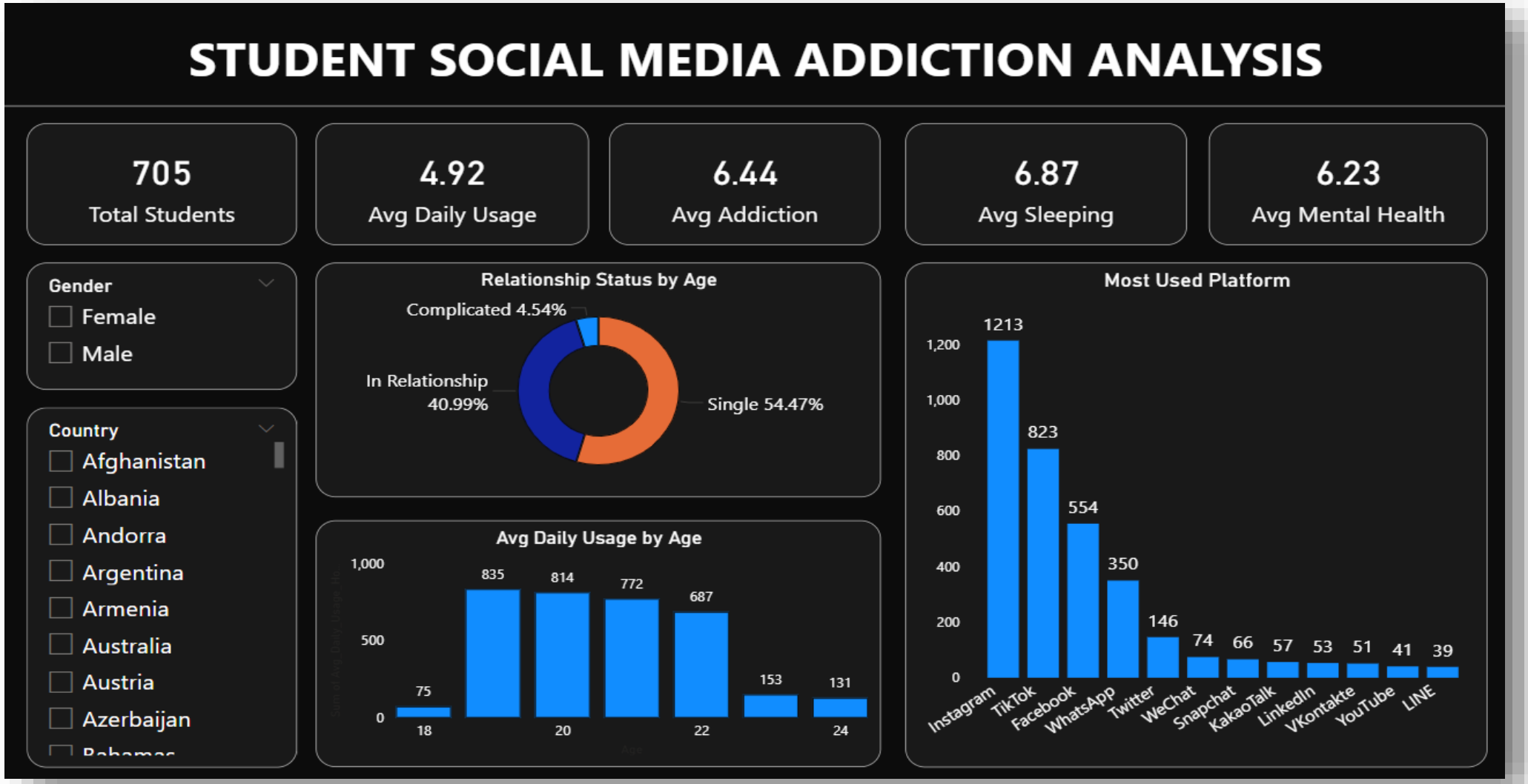
This report examines whether social media usage affects students' academic performance. The visualization identifies the proportion of students who reported that social media usage impacts their studies.

- **Sleep Pattern Analysis**

Sleep hours per night are analyzed in relation to social media usage. The report indicates that students with higher social media engagement tend to have fewer sleep hours, suggesting a possible relationship between digital activity and sleep patterns.

Overall, the reports generated through the dashboard provide clear insights into how social media usage influences different aspects of students' lives, helping stakeholders better understand behavioral trends and potential areas of concern.

RESULT:


STUDENT SOCIAL MEDIA ADDICTION ANALYSIS
705
Total Students
4.92
Avg Daily Usage
6.44
Avg Addiction
6.87
Avg Sleeping
6.23
Avg Mental Health
Gender
Female
Male
Country
Afghanistan
Albania
Andorra
Argentina
Armenia
Australia
Austria
Azerbaijan
Relationship Status by Age
Complicated 4.54%
In Relationship 40.99%
Single 54.47%
Avg Daily Usage by Age
1,000
500
0
75
835
814
772
687
153
131
18
20
22
24
Most Used Platform
1,200
1,000
800
600
400
200
0
1213
823
554
350
146
74
66
57
53
51
41
39
Instagram
TikTok
Facebook
WhatsApp
Twitter
WeChat
Snapchat
KakaoTalk
LinkedIn
VKontakte
YouTube
LINE

CONCLUSION:

- The **Student Social Media Addiction Analysis** project helps to understand how students use social media and how it affects their daily life. By analyzing data such as **daily usage hours, addiction score, mental health score, sleep hours, and academic performance**, useful patterns and relationships were identified.
- The Power BI dashboard presents the data through different visualizations like **bar charts, pie charts, maps, KPI cards, and gauge charts**, which makes the analysis easy to understand. From the analysis, it is observed that students who spend more time on social media tend to have **higher addiction scores and fewer sleeping hours**, which may impact their academic performance and overall well-being.
- The dashboard also highlights the **most used social media platforms** and shows how social media usage varies among students based on different factors such as gender, academic level, and country.

Overall, this project demonstrates how **data visualization and analytics can help transform raw data into meaningful insights**, helping educators and researchers better understand students' social media usage patterns.

FUTURE SCOPE:

- The **Student Social Media Addiction Analysis** project can be further improved by expanding the dataset and adding more advanced analytical features. A larger dataset with more students from different regions and academic backgrounds can provide more accurate and detailed insights into social media usage patterns.
- In the future, additional factors such as **screen time trends, academic grades, and psychological factors** can be included to better understand the relationship between social media usage and student performance. This will help in performing deeper analysis of students' digital behavior.
- The dashboard can also be enhanced by adding **predictive analytics and machine learning models** to forecast addiction levels or predict how social media usage might affect academic performance and mental health.
- Furthermore, the project can be integrated with **real-time data sources** to monitor social media usage patterns continuously. This can help educators and researchers make better decisions and promote awareness about balanced and healthy use of social media among students.

REFERENCES:

Dataset: Student Social Media Addiction Dataset Retrieved from Kaggle Data Repository.

Kaggle:https://www.kaggle.com/datasets/zahranusratt/student-social-media-addiction-analysis-dataset

GitHub Link: https://github.com/shakeer456/STUDENT-SOCIAL-MEDIA-ADDICTION-ANALYSIS

LinkedIn: https://www.linkedin.com/posts/shakeer-shaik-508958235_datascienece-dataanalytics-powerbi-activity-7437730063737028609-MBbq?utm_source=share&utm_medium=member_desktop&rcm=ACoAADqyCdgBTycPAYzc6770UZAtYoHWgcz-ZSg

Thank You